EXPLAINABILITY OF VISION TRANSFORMERS

Inherently Explainable and Task-specific Methods, Evaluation, Tools and Outlook

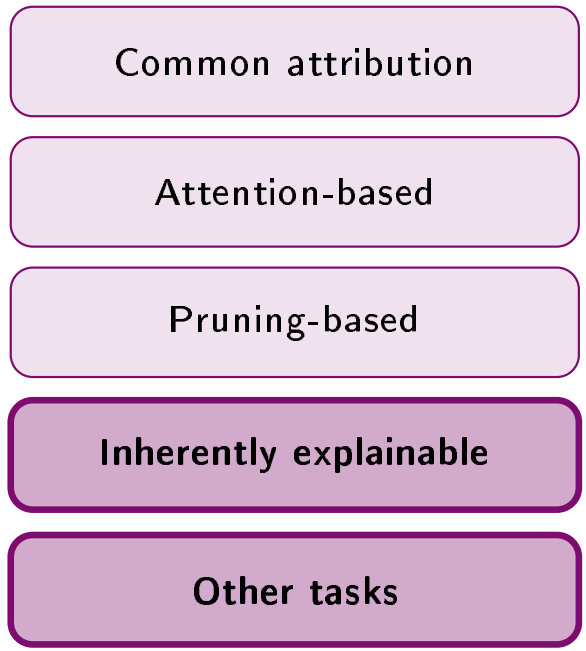
Sections 3.4–8 of Kashefi et al., *Explainability of Vision Transformers: A Comprehensive Review and New Perspectives* (arXiv:2311.06786, 2023)

[Your name] [Teammates]

University of Science - VNUHCM

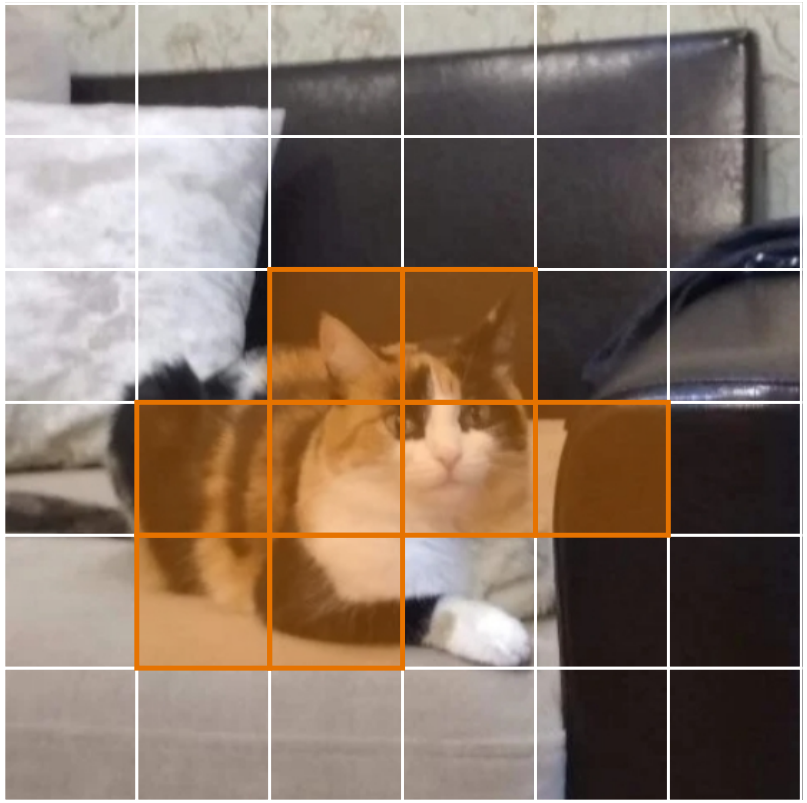
Outline of this part

3.4 Inherently explainable methods
3.5 Explainability of other tasks
4 Explanation evaluation
5–6 Tools and datasets
7–8 Discussion and conclusion

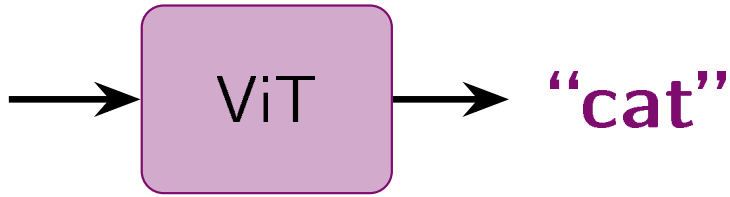


Taxonomy of the survey (Fig. 2): bold = this part

Running example: a cat on a sofa



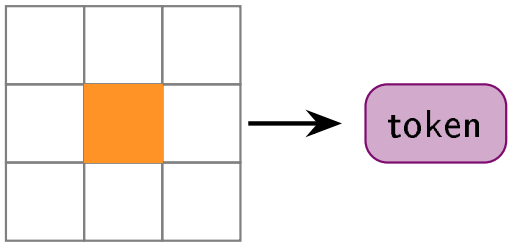
$6 \times 6 = 36$ patches: **8 cat** + 28 sofa



Which patches did the model really need to say “cat”?

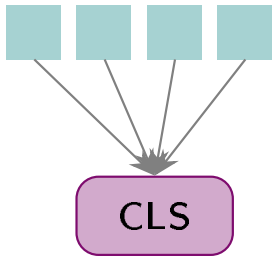
Photo: Kafun, CC BY-SA 4.0, Wikimedia Commons. A real ViT uses $14 \times 14 = 196$ patches; we draw 36.

Five key terms in pictures



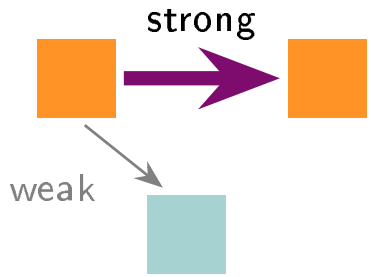
Patch / token

a piece of the image



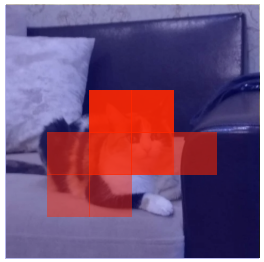
Class token

gathers all tokens, answers



Attention

one token looks at another

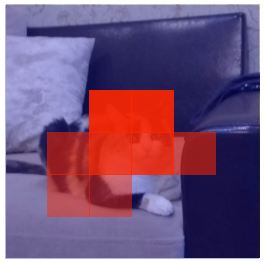


Saliency map

bright = important



model uses

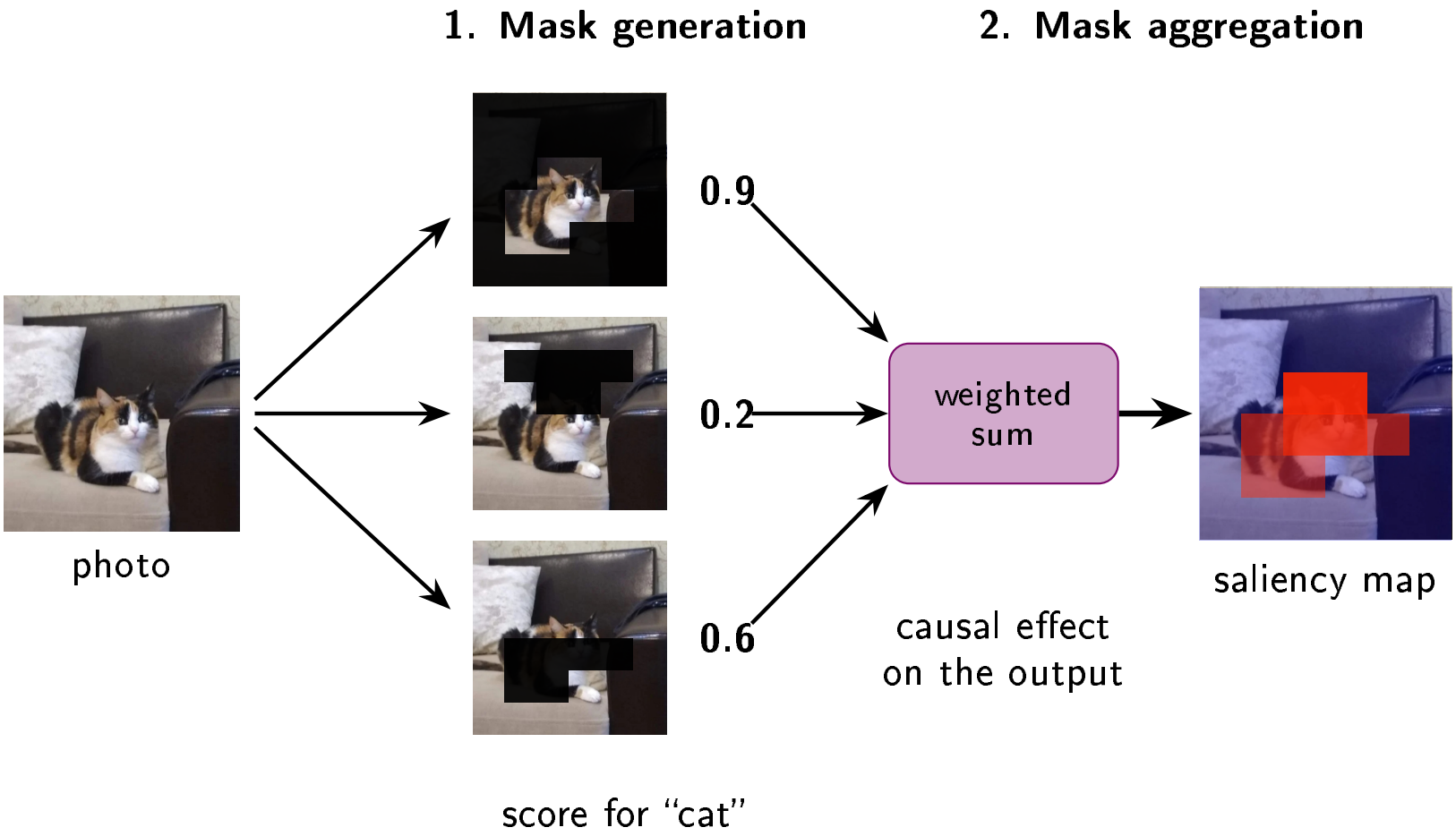


human expects

Faithful vs. plausible

3.4 Inherently explainable: ViT-CX

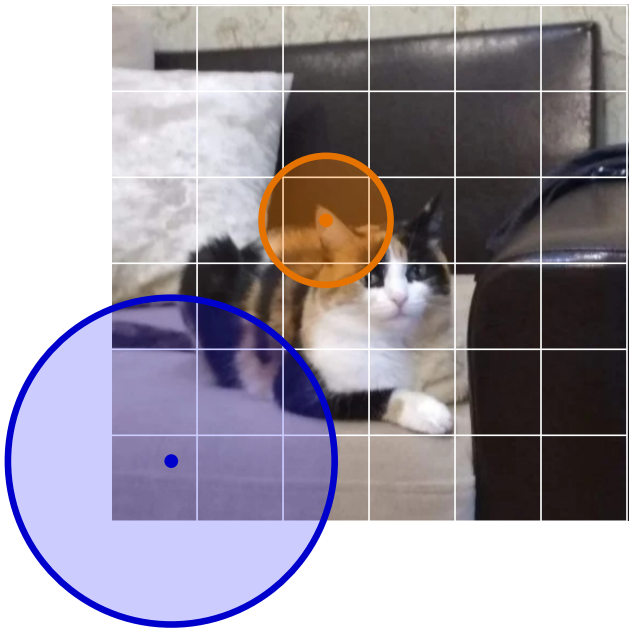
Self-explaining models: trees, rules, sparsity (less accurate)



Cat drawings: our own illustration, not from the paper.

Patch interactions & ViT-NeT

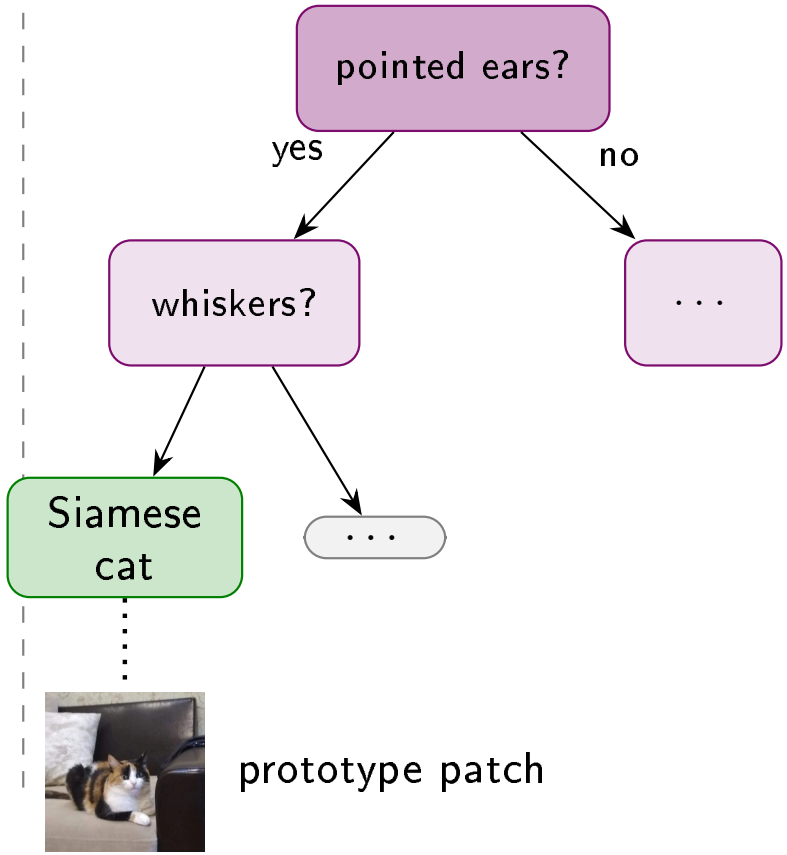
Patch interactions



object patch: small window

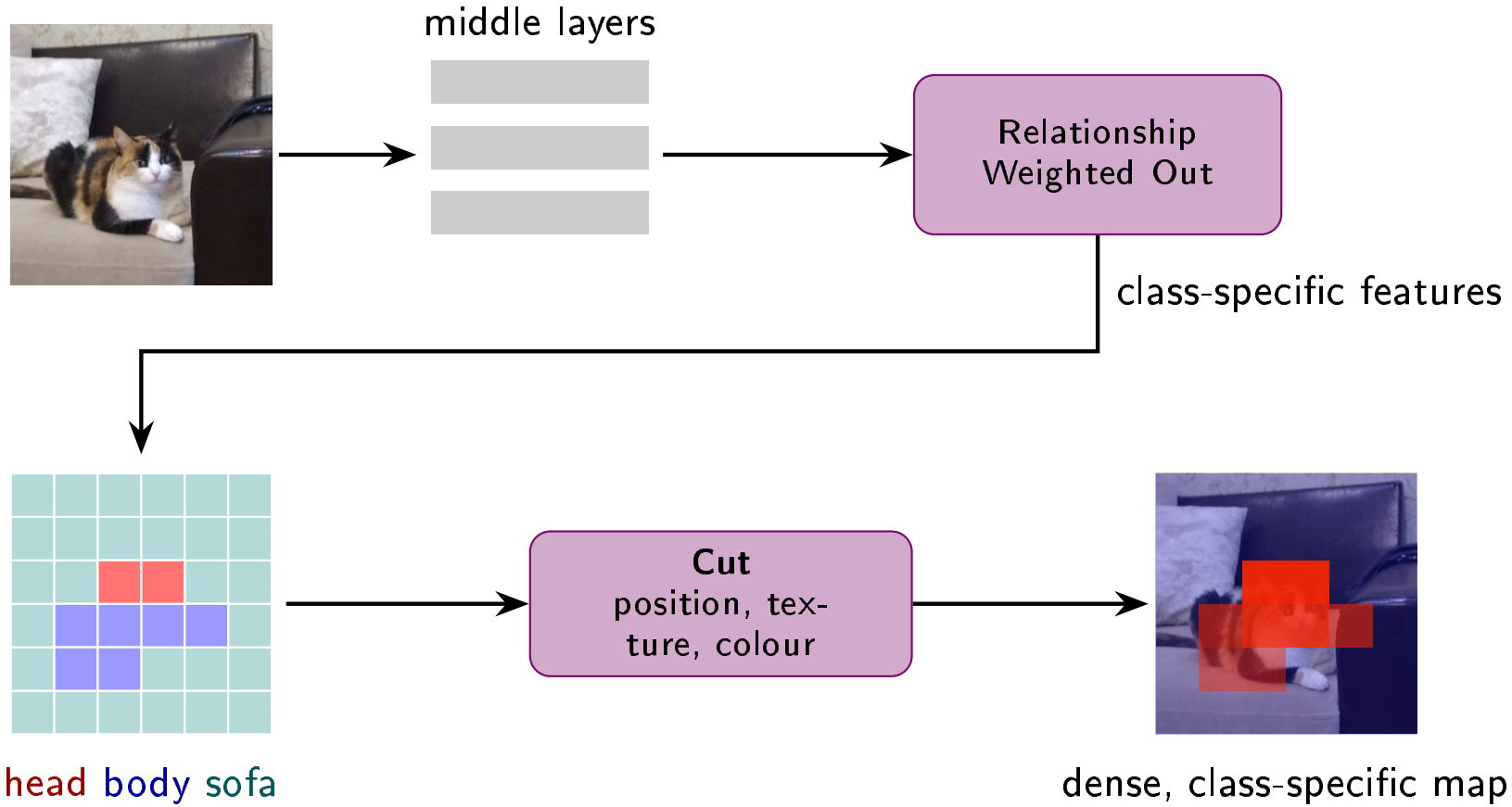
background patch: large window

ViT-NeT: tree decoder



Cat drawings: our own illustration, not from the paper.

R-Cut

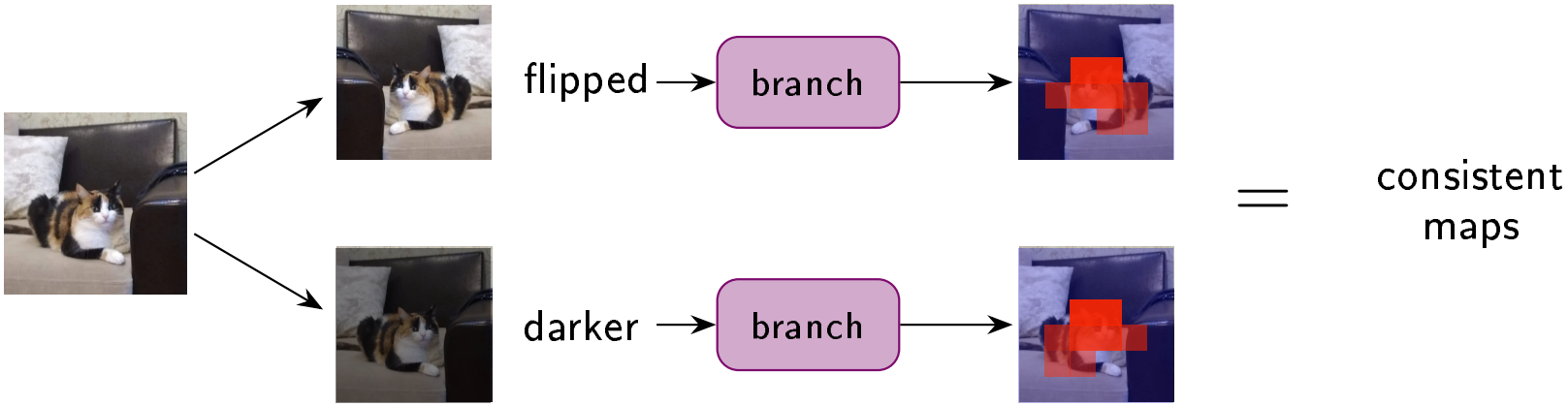


Evaluated with the **IoU test** (Section 4)

Cat drawings: our own illustration, not from the paper.

3.5 Other tasks: eX-ViT and DINO

eX-ViT: weakly supervised segmentation



DINO: no labels

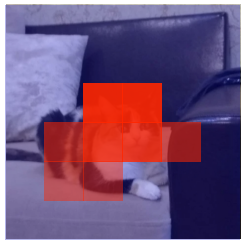


Cat drawings: our own illustration, not from the paper.

Generic Attention-model, ATMAN, Concept-Transformer

Generic

bi-modal, co-attention



patches **and** words

What is on the **sofa** ?

↓

"cat"

ATMAN

perturbs attention



weaken the face

score before

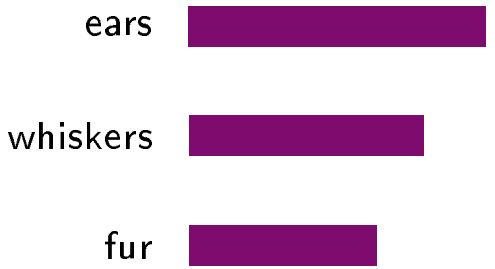


score after



Concept-Transformer

attention over concepts



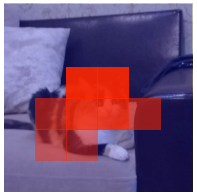
concepts, not pixels

↓

"cat"

Cat drawings: our own illustration, not from the paper.

4 Explanation evaluation

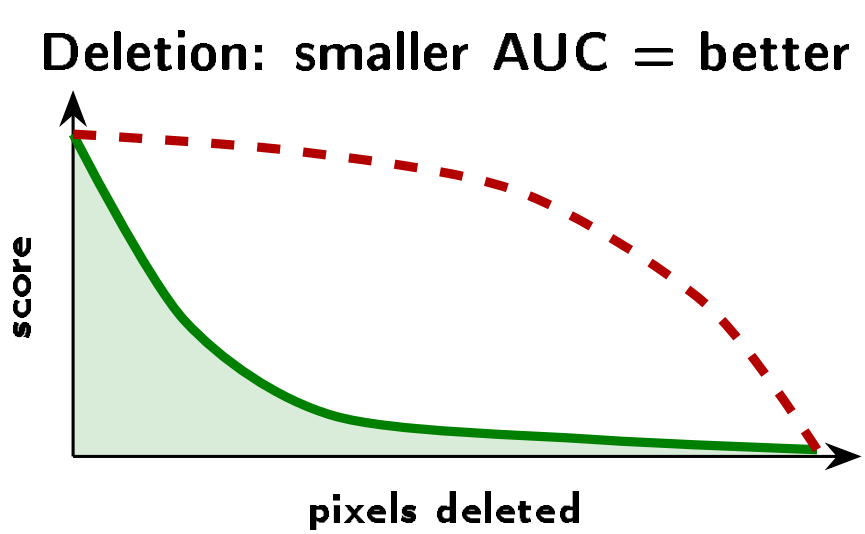


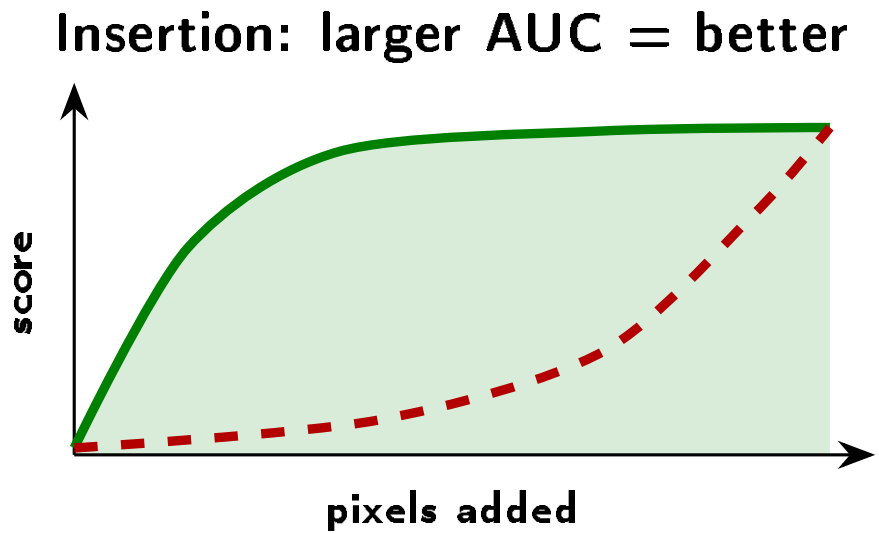
?

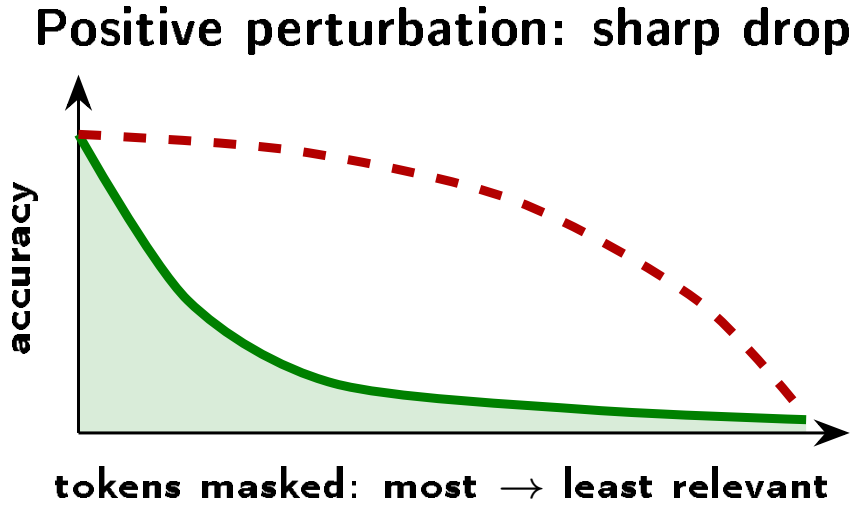
is this map right? → **Evaluative criteria**

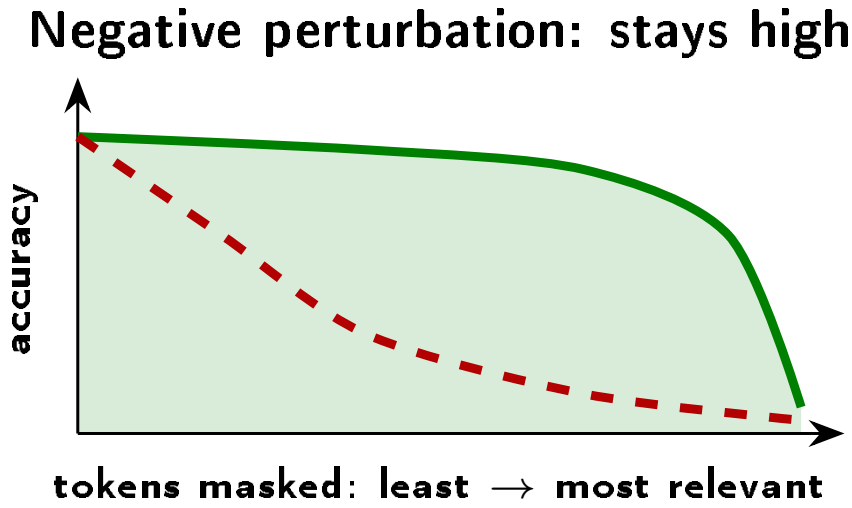
- Deletion / Insertion
- Effective Complexity
- Faithfulness
- (In)fidelity
- Sensitivity
- Sensitivity-n
- Perturbation tests
- IoU test
- (Grid) Pointing Game
- Segmentation tests

Deletion / Insertion and Perturbation tests





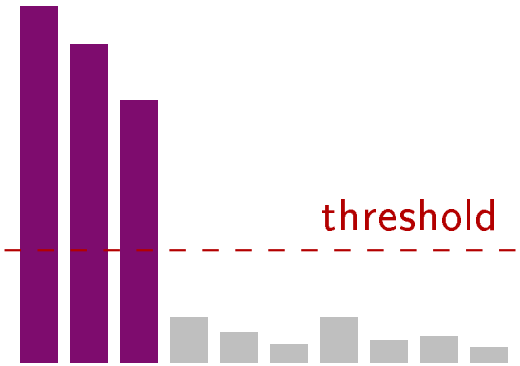




good map (cat first) bad map (sofa first)

Effective Complexity, Faithfulness, (In)fidelity

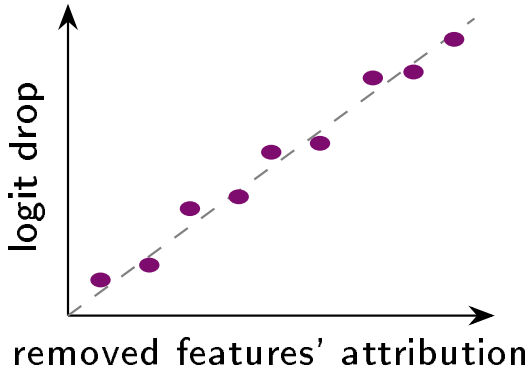
Effective Complexity



count values above it

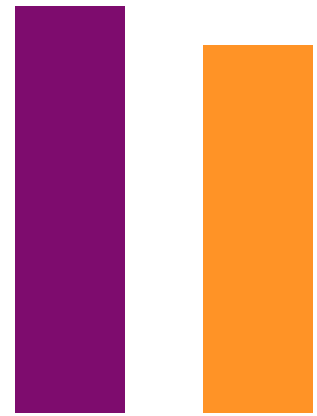
EC = 3: simple

Faithfulness



correlation in $[-1, 1]$

(In)fidelity



real change

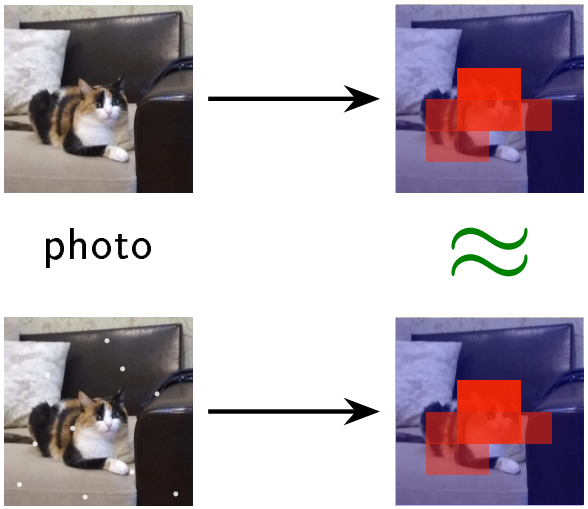
map predicts

close = low infidelity

Sensitivity and Sensitivity-n

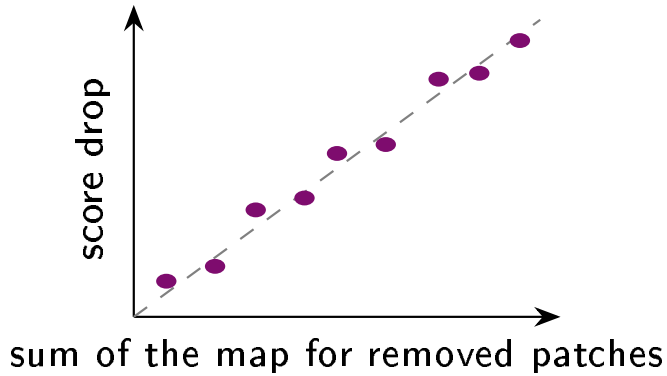
Sensitivity

tiny noise $\Rightarrow$ same map?



Sensitivity-n

remove n random patches

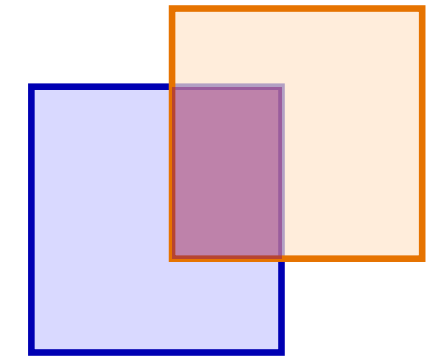


high correlation = accurate

Cat drawings: our own illustration, not from the paper.

IoU, Pointing Game and Segmentation tests

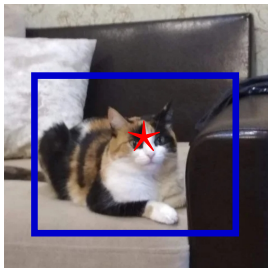
IoU test



true box map > 0.2

$$\text{IoU} = \frac{\cap}{\cup}$$

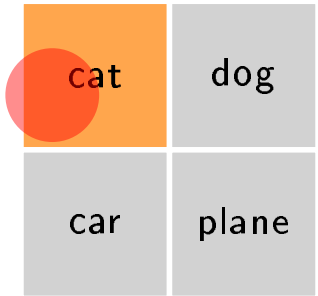
Pointing Game



brightest point
in the box = hit

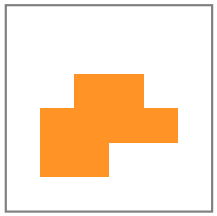
$$\frac{\#\text{hits}}{\#\text{hits} + \#\text{misses}}$$

Grid Pointing Game

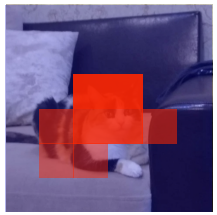


attribution
should land
on the cat cell

Segmentation tests



truth

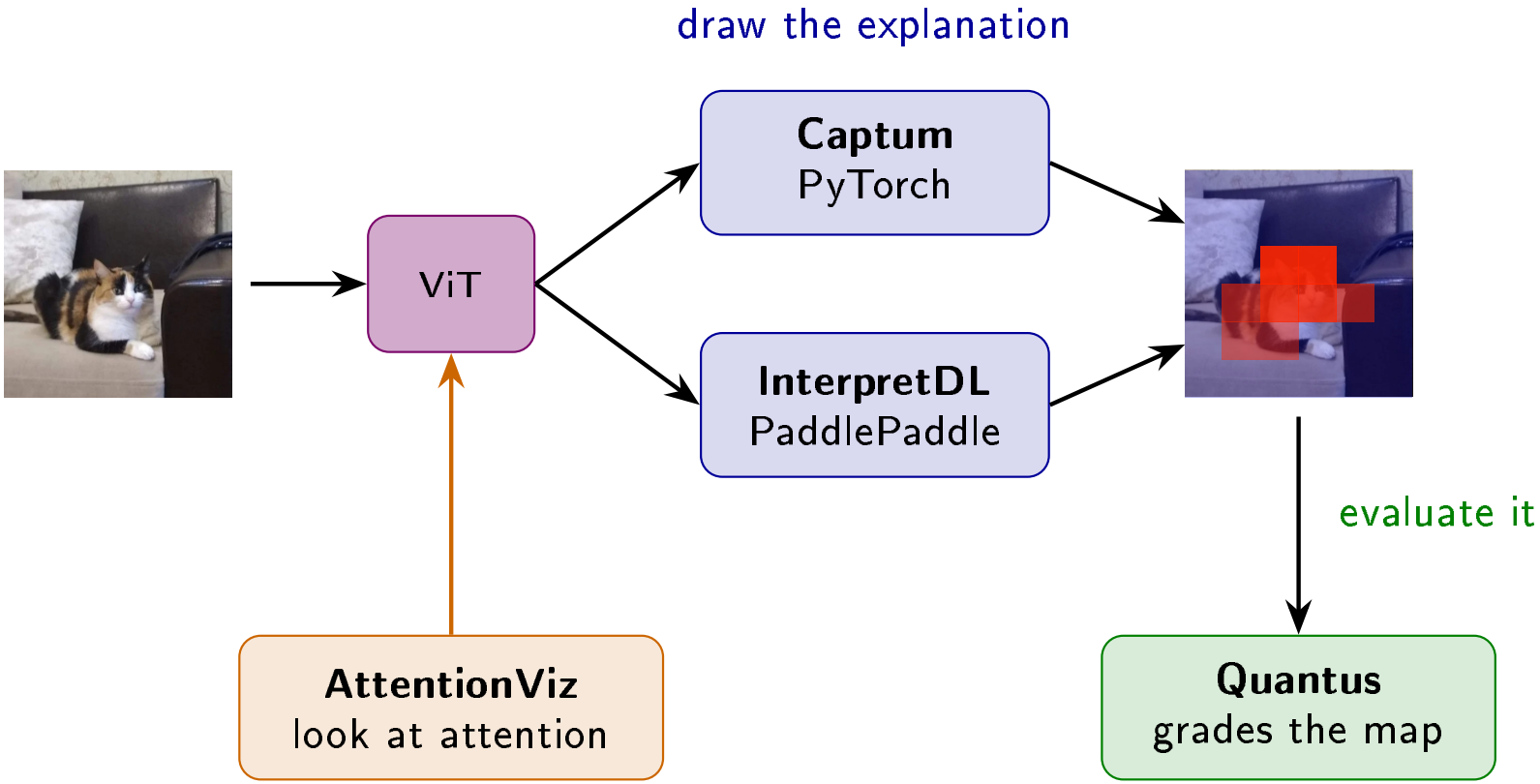


map

acc., mIoU,
mAP, mF1

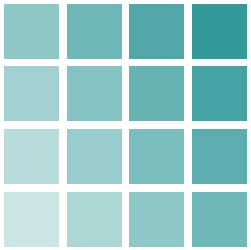
Cat drawings: our own illustration, not from the paper.

5 Tools and frameworks



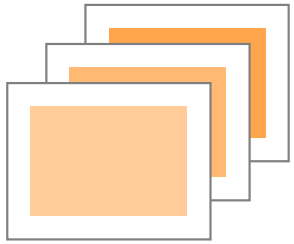
Cat drawings: our own illustration, not from the paper.

6 Datasets



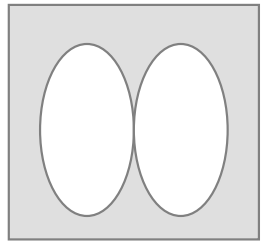
CIFAR-10

60,000 small images



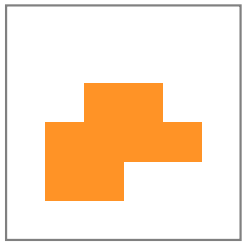
ImageNet

millions of images



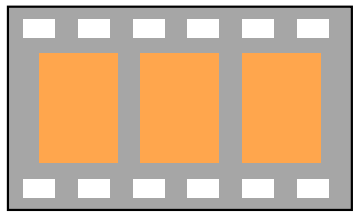
COVID-QU-Ex

33,920 chest X-rays



ImageNet-Segmentation

for segmentation tests

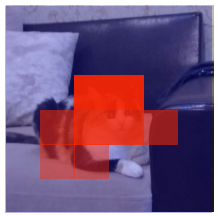


Kinetics-400

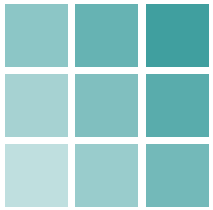
video classification

7 Discussion and future works

Attribution vs. visualization



local: which pixels?



global: what was learned?
less explored

Attention alone is not enough



raw attention

$\neq$ "cat"

Beyond classification

classi-fication

multi-modal

genera-tive

the last two: nascent

Use explanations for

debugging

improving

fairness

reliability

Cat drawings: our own illustration, not from the paper.

8 Conclusion

An overview of explainability techniques for ViT

Common attribution | Attention-based | Pruning-based | Inherently explainable | Other tasks

5 groups: by motivation, structure, application scenario

Evaluation criteria | **Tools** and frameworks | **Open issues** and future directions

Goal: a better understanding of the inner mechanisms of ViTs

References I


R. Kashefi, L. Barekatain, M. Sabokrou, F. Aghaeipoor. Explainability of vision transformers: A comprehensive review and new perspectives. *arXiv:2311.06786*, 2023.


H. Chefer, S. Gur, L. Wolf. Transformer interpretability beyond attention visualization. *CVPR*, 2021.


W. Xie et al. ViT-CX: Causal explanation of vision transformers. *arXiv:2211.03064*, 2022.


J. Ma et al. Visualizing and understanding patch interactions in vision transformer. *IEEE TNNLS*, 2023.


S. Kim et al. ViT-NeT: Interpretable vision transformers with neural tree decoder. *ICML*, 2022.


Y. Niu et al. R-Cut: Enhancing explainability in vision transformers with relationship weighted out and cut. *arXiv:2307.09050*, 2023.


L. Yu et al. eX-ViT: A novel explainable vision transformer for weakly supervised semantic segmentation. *Pattern Recognition*, 2023.


M. Caron et al. Emerging properties in self-supervised vision transformers. *ICCV*, 2021.


H. Chefer, S. Gur, L. Wolf. Generic attention-model explainability for interpreting bi-modal and encoder-decoder transformers. *ICCV*, 2021.


M. Deb et al. AtMan: Understanding transformer predictions through memory efficient attention manipulation. *arXiv:2301.08110*, 2023.


M. Rigotti et al. Attention-based interpretability with concept transformers. *ICLR*, 2022.

References II


V. Petsiuk, A. Das, K. Saenko. RISE: Randomized input sampling for explanation of black-box models. *arXiv:1806.07421*, 2018.


C. Rudin. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 2019.


N. Kokhlikyan et al. Captum: A unified and generic model interpretability library for PyTorch. *arXiv:2009.07896*, 2020.


X. Li et al. InterpretDL: Explaining deep models in PaddlePaddle. *JMLR*, 2022.


C. Yeh et al. AttentionViz: A global view of transformer attention. *arXiv:2305.03210*, 2023.


A. Hedström et al. Quantus: An explainable AI toolkit for responsible evaluation of neural network explanations and beyond. *JMLR*, 2023.

Thank you!

Questions and discussion